

一、逻辑与推理

有一个保险柜，密码是由 NOMONEY 组成的 7 位数字。每个字母对应的数字隐藏在下面的 10 进制加法算式中，相同的字母对应于相同的数字，不同的字母对应于不同的数字，数字范围 0-9，并且已知密码中没有 4。请解出密码。(20 分)

$$\begin{array}{r}
 \text{G I V E} \\
 + \text{M O R E} \\
 \hline
 \text{M O N E Y}
 \end{array}$$

二、C/C++ 语言

- 请写一个函数，实现如下功能：统计出一个 32 位整数中第一个非 0 位之前 0 位的个数。如：输入 0x01 应返回 31。输入 0x10000，返回 15。(机器字长 32 位)。(5 分)
 - 函数原型: `int CountLeadingZero(unsigned int vp_Value);`
- 请用 C 语言写一个函数，实现如下功能：输入年、月、日，返回这一天是这一年的第几天？请注意算法的效率（空间/时间）。(20 分)

方法原型: `int dayOfYear(int y, int m, int d)`
- 什么样的代码是优美的？哪些好的习惯有助于提高代码质量？(5 分)
- C 和 C++ 有哪些主要区别？(5 分)
- 请从下面二个问题任选一个(15 分)
 - 请描述 C++ 中各种智能指针 (`shared_ptr`、`auto_ptr`、`unique_ptr`、`weak_ptr`) 的适用场景及注意事项。
 - 请简述一下 Linux 中几种多路复用机制 (`select`/`poll`/`epoll`) 的优缺点对比。
- 翻译：(10 分)

When developing embedded software, you must consider the following:

- Understand the default compilation tool behavior and the target environment so that you appreciate the steps necessary to move from a debug or development build to a fully standalone production version of the application.
- Some C library functionality executes by using debug environment resources. If used, you must reimplement this functionality to make use of target hardware.
- The toolchain has no inherent knowledge of the memory map of any given target. You must tailor the image memory map to the memory layout of the target hardware.
- An embedded application must perform some initialization, such as stack and heap initialization, before the main application can be run. A complete initialization sequence requires code that you implement in addition to the Arm Compiler C library initialization routines.

- 找出下面代码中所有的错误以及不合理的地方，并请说明原因。(20 分)


```

1: #include <vector.h>
2: class A
3: {
4:     public:
5:         A() {m_iCount = 0;}
6:         ~A(){}
7:     private:
8:         int m_iCount;
9: };
10: class B: public A
11: {
12:     public:
13:         B():A() {m_iCount = 1;}
14:         ~B(){}
15: };
16:
17: void main(int argc, char** argv)
18: {
19:     A* a = new B();
20:     FILE* f;
21:     int n;
22:     if (argc > 1)
23:         n = argv[0];
24:     char* stuff = new char[n];
25:     f = fopen("c:/a/b/c/aa.txt", "r");
26:     fread(f, 1, n, stuff);
27:     n = strlen(stuff);
28:     if (n < 10)
29:     {
30:         return -1;
31:     }
32:     memset(stuff, 0x20, 10);
33:     fwrite(f, 1, n, stuff);
34:     delete stuff;
35:     delete a;
36:
37:     cout << stuff;
38:     return 0;
39: }

```